

SKETCHABLE INFINITY CATEGORIES

David Martínez-Carpena

Carles Casacuberta Javier J. Gutiérrez

Categorical Logic and Higher Categories

Manchester, 16-19 December 2024



UNIVERSITAT DE
BARCELONA

Sketches

A **sketch** Σ is a triple (A, L, C) of a small category A , a set L of cones and a set C of cocones. If $C = \emptyset$, Σ is a **limit sketch**.

A **model** of a sketch in a category K is a functor $F : A \rightarrow K$ which sends cones of L to limits and cocones of C to colimits.

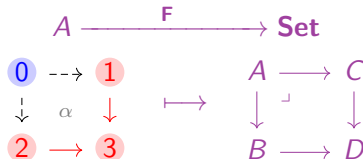
$\text{Mod}(\Sigma, K) := \infty\text{-category of models of } \Sigma \text{ in } K$

$\text{Mod}(\Sigma) := \infty\text{-category of models of } \Sigma \text{ in } \mathbf{Set}$

Example. Let A be the small category generated by the square α :

Observe that α is a cone
with **apex** and **diagram**

Then $(A, \{\alpha\}, \emptyset)$ is a sketch 🍷



A model F of the sketch $(A, \{\alpha\}, \emptyset)$ is a pullback of sets 🍷

Representation theorem

Limit-sketchable $\iff$ Equivalent to $\text{Mod}(\Sigma)$ for a limit sketch Σ

Sketchable $\iff$ Equivalent to $\text{Mod}(\Sigma)$ for a sketch Σ

Theorem (Gabriel and Ulmer 1971)

A category is presentable if and only if it is limit-sketchable.

Theorem (Lair 1981)

A category is accessible if and only if it is sketchable.

Goal

Generalize these results to ∞ -categories

Higher sketches (Joyal 2008)

A **sketch** Σ is a triple $(\mathcal{A}, L, C)$ of a small ∞ -category $\mathcal{A}$, a set L of cones and a set C of cocones. If $C = \emptyset$, Σ is a **limit sketch**.

A **model** of a sketch in a category $\mathcal{K}$ is a functor $F : \mathcal{A} \rightarrow \mathcal{K}$ which sends cones of L to limits and cocones of C to colimits.

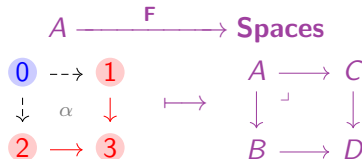
$\text{Mod}(\Sigma, \mathcal{K}) := \infty\text{-category of models of } \Sigma \text{ in } \mathcal{K}$

$\text{Mod}(\Sigma) := \infty\text{-category of models of } \Sigma \text{ in } \mathbf{Spaces}$

Example. Let $\mathcal{A}$ be the nerve of the category generated by α :

Observe that α is a cone
with apex and diagram

Then $(\mathcal{A}, \{\alpha\}, \emptyset)$ is a sketch 🍷



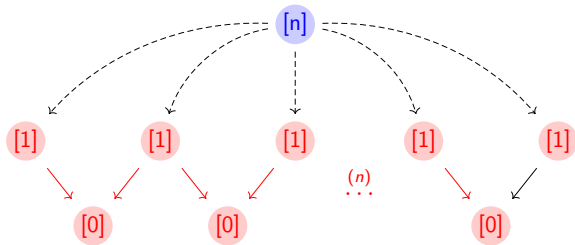
A model F of the sketch $(\mathcal{A}, \{\alpha\}, \emptyset)$ is a homotopy pullback 🍷

Examples

- **Algebraic theories** (Rosicky 2007): Monoid objects (A_∞ -spaces), commutative monoid objects (E_∞ -spaces), group objects (∞ -groups), R-modules, ...
- **Algebraic patterns** (Chu and Haugseng 2021): Category objects (Segal spaces), operad objects (Segal dendroidal spaces), ...
- Spectrum object (Spectra)
- Univalent category objects (Complete Segal spaces) and univalent operad objects (Complete Segal dendroidal spaces)
- ∞ -sheaves $\simeq$ hypercomplete ∞ -topos

Examples: Category objects

Let $\mathcal{K}$ be a complete ∞ -category, and α_n be the cone over Δ^{op} with **apex** and **diagram** for all $n \in \mathbb{N}$:



Then $\Sigma = (\Delta^{\text{op}}, \{\alpha_n \mid n \in \mathbb{N}\})$ is a limit sketch, and a model $F : \Delta^{\text{op}} \rightarrow \mathcal{K}$ is a simplicial object in $\mathcal{K}$ such that

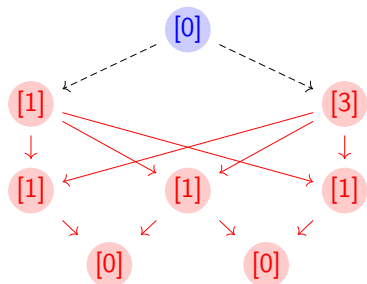
$$F_n \xrightarrow{\sim} F_1 \times_{F_0} F_1 \times_{F_0} \overset{(n)}{\cdots} \times_{F_0} F_1. \quad (\text{Segal condition})$$

$$\text{Mod}(\Sigma, \mathcal{K}) \simeq \text{Category objects on } \mathcal{K}$$

$$\text{Mod}(\Sigma) \simeq \text{Segal spaces}$$

Examples: Univalent category objects

Let $\mathcal{L}_S$ be the set of cones of the previous example, and β be the cone over Δ^{op} with **apex** and **diagram**:



Then $\Sigma = (\Delta^{\text{op}}, \mathcal{L}_S \cup \{\beta\})$ is a limit sketch, and a model $F : \mathcal{A} \rightarrow \mathcal{K}$ is a category object in $\mathcal{K}$ such that

$$\begin{array}{ccc}
 F_0 & \xrightarrow{\quad} & F_3 \\
 \downarrow & \lrcorner & \downarrow \\
 F_1 & \longrightarrow & F_1 \times_{F_0}^{d_1, d_1} F_1 \times_{F_0}^{d_0, d_0} F_1
 \end{array}$$

$\text{Mod}(\Sigma, \mathcal{K}) \simeq$ **Univalent category objects on $\mathcal{K}$**

$\text{Mod}(\Sigma) \simeq$ **Complete Segal spaces**

Higher representation theorem

Theorem. An ∞ -category is:

- (a) presentable if and only if it is limit-sketchable,
- (b) and accessible if and only if it is sketchable.

- ✓ Presentability definitions of Joyal and Lurie are equivalent.
- ✓ Can be used to prove that many ∞ -categories are presentable.
- ✓ The result can be extended to models of sketches over any presentable ∞ -category.
- ❓ Example of a sketch with ∞ -category of models that is accessible but neither presentable nor 1-categorical.

Thank you for listening!

-  Joyal, André (2008). *The Theory of Quasi-Categories and its Applications*. Barcelona: Lectures at CRM.
-  Rosicky, Jiří (2007). “On homotopy varieties”. *Advances in Mathematics* 214.2.
-  Chu, Hongyi and Rune Haugseng (2021). “Homotopy-coherent algebra via Segal conditions”. *Advances in Mathematics* 385.

SKETCHABLE INFINITY CATEGORIES

David Martínez-Carpena

Carles Casacuberta Javier J. Gutiérrez

Categorical Logic and Higher Categories

Manchester, 16-19 December 2024



UNIVERSITAT DE
BARCELONA